04/12/24 M/S

USN | | | | | | | | | | | **[23CSBS301]**

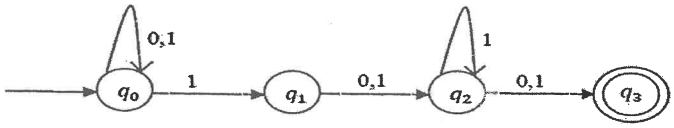
THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

FORMAL LANGUAGE AND AUTOMATA THEORY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5x2=10**
1.	**a.**	Define (i) Finite Automaton (FA) (ii)Transition diagram	L1	CO1	**02**
	b.	Draw the Transition table for given the finite automaton 	L1	CO2	**02**
	c.	State the pumping lemma for CFLs.	L1	CO3	**02**
	d.	List out the components of Turing Machine	L1	CO4	**02**
	e.	Define NP Complete	L1	CO5	**02**
		PART - B			**6x5=30**
2.	**a.**	Explain the difference between DFA and NFA	L2	CO1	**05**
		OR			
	b.	Draw DFA accepting all strings over w $\in(0,1)^*$ which contains "0111" as a sub string	L3	CO1	**05**
3.	**a.**	Explain the procedure to Convert from NFA to DFA	L2	CO1	**05**
		OR			
	b.	Construct NFA for the given Regular Expression 01(0+1)	L6	CO2	**05**
4.	**a.**	Obtain grammar to generate string consisting of any number of a's and b's with at least one b.	L3	CO2	**05**
		OR			
	b.	Obtain a grammar to generate the following language: $L = \{ w \mid n_a(w) > n_b(w) \}$	L3	CO2	**05**
5.	**a.**	Define Push Down Automata (PDA) Explain in detail.	L3	CO3	**05**
		OR			
	b.	Design a PDA for accepting a language $\{0^n \mid n>=1\}$.	L6	CO3	**05**
6.	**a.**	Explain the basic model of Turing machine	L2	CO4	**05**
		OR			
	b.	Explain Recursive Language in Turing Machine.	L2	CO4	**05**
7.	**a.**	State and prove Church-Turing Thesis.	L5	CO5	**05**
		OR			
	b.	Prove that universal language L_u is a recursively enumerable language but not recursive.	L5	CO5	**05**

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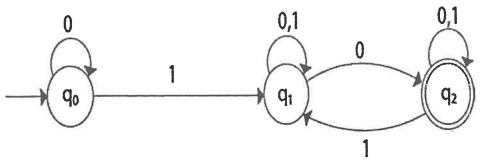
		PART - C			6x10=60
8.	**a.**	Construct DFA accepting the set of all strings beginning with 101 such that { w is the length of the string ,|w| >= 6}	L6	CO1	**10**
		OR			
	b.	Convert from NFA to DFA for the given NFA transition diagram 	L1	CO1	**10**
9.	**a.**	Design DFA for the given expression. a(ab)*aa.	L6	CO2	**10**
		OR			
	b.	Construct NFA for the given Regular Expression $(1+0)(0 + 10)^*$	L6	CO2	**10**
10.	**a.**	Convert the given CFG to CNF. Consider the given grammar G1: $E \rightarrow E+T \mid T$ $T \rightarrow T*F \mid F$ $F \rightarrow (E) \mid a$	L1	CO2	**10**
		OR			
	b.	Convert the given CFG to CNF. Consider the given grammar G1: $S \rightarrow a \mid aA \mid B$ $A \rightarrow aBB \mid \varepsilon$ $B \rightarrow Aa \mid b$	L6	CO2	**10**
11.	**a.**	Construct pushdown automata for the given languages. Acceptance either by empty stack or by final state. $\{ a^n b^m c^m \mid m, n \in N \}$	L6	CO3	**10**
		OR			
	b.	Convert Context Free Grammar into Push Down automata $E \rightarrow E+E$ $E \rightarrow E*E$ $E \rightarrow a$	L6	CO3	**10**
12.	**a.**	Construct a TM that accepts input string does not contain a sub string with 'aba' where all input made of the combination of {a,b}	L6	CO4	**10**
		OR			
	b.	Explain the closure properties of Recursive language in Turing Machine	L2	CO4	**10**
13.	**a.**	Explain the following problems **i.** NP Hard **ii.** NP Complete	L2	CO5	**10**
		OR			
	b.	Consider the correspondence system as given below A = (b, bab^3, ba) and B = (b^3, ba, a). The input set is Σ = {0, 1}. Find the solution.	L3	CO5	**10**